
ST-Veto: Spatio-Temporal Token Veto for Diffusion MLLMs via Taylor Prediction and Visual Grounding

Keuntae Kim^{1*} Beomseok Lee^{2*} Hyunwoo Kim^{3*} Yong Suk Choi^{1†}

Abstract

Vision Language Models (VLMs) achieve strong reasoning with Chain-of-Thought (CoT) prompting but incur high sequential-generation cost, error accumulation, and limited self-correction. Diffusion Multimodal Large Language Models (dM-LLMs) unmask tokens in an order-agnostic process, improving efficiency and enabling iterative refinement, yet their reasoning and how to enhance it remain underexplored. We propose a training-free method, Spatio-Temporal Token Veto (ST-Veto), which leverages the ability to observe all token positions at each diffusion step. Rather than relying only on current-step confidence, ST-Veto vetoes temporally unstable tokens via second-order Taylor prediction of confidence dynamics and filters weakly grounded tokens using image-attention mass, swapping them with safer candidates. Across multiple dM-LLMs and multimodal reasoning benchmarks, ST-Veto consistently outperforms standard decoding policies and prior VLM reasoning methods, improving accuracy by up to 9% with no additional training or generation cost. Analyses show that ST-Veto steers generation toward higher-confidence, better-grounded paths.

1. Introduction

Large Language Models (LLMs) have established a strong foundation in NLP tasks through next-token prediction. Building upon this, Vision Language Models (VLMs), which integrate LLMs as backbones with vision encoders, have achieved strong results in the multimodal domain (Alayrac et al., 2022; Li et al., 2023b;a; Wang et al.,

2024; Zhang et al., 2024a). Central to the success of both LLMs and VLMs is the Auto-Regressive (AR) mechanism, which generates tokens sequentially. Furthermore, conditioning on previously generated tokens, such as via Chain-of-Thought (CoT) prompting, has substantially enhanced the reasoning capabilities of AR models (Wei et al., 2022; Zhang et al., 2022; Wang et al., 2022).

However, the sequential nature of AR models incurs high latency and inference costs. A critical limitation is the difficulty of self-correction: because the model continually consumes its own generated tokens as input, errors can easily propagate and amplify hallucinations (Ji et al., 2023; Pan et al., 2023). Recent diffusion-based language models instead generate tokens through order-agnostic denoising and iterative refinement (Li et al., 2022; Gong et al., 2022; Sahoo et al., 2024b; Nie et al., 2025). Diffusion Large Language Models (dLLMs) and diffusion Multimodal Large Language Models (dM-LLMs) leverage these properties to improve efficiency and enable bidirectional context modeling, facilitating capabilities such as self-correction (Li et al., 2025; Yang et al., 2025; You et al., 2025).

Despite this potential, d(M)LLMs remain in the nascent stages of research and have not yet reached the maturity of AR models, particularly regarding reasoning. Prior works have primarily emphasized efficiency by restricting evaluation to relatively simple tasks with short generation lengths (e.g., 16 or 32 tokens) (Li et al., 2025; Yang et al., 2025). As a result, it remains unclear whether dM-LLMs can sustain non-trivial reasoning under long-generation settings or how they internally perform inference. For instance, whether dM-LLMs can generate coherent and faithful rationales when given sufficient generation length has received limited attention. Moreover, since standard CoT prompting does not directly translate to the non-sequential generation paradigm, a key open question is how to improve reasoning reliability in diffusion models (Sahoo et al., 2025).

In this study, we analyze the reasoning capabilities of dM-LLMs and propose a method to enhance them. We evaluate two state-of-the-art dM-LLMs, LaViDa-Reason (Li et al., 2025) and MMaDA-MixCoT (Yang et al., 2025), on benchmarks requiring multimodal understanding and reasoning (Chen et al., 2024; Liu et al., 2024; Lu et al.,

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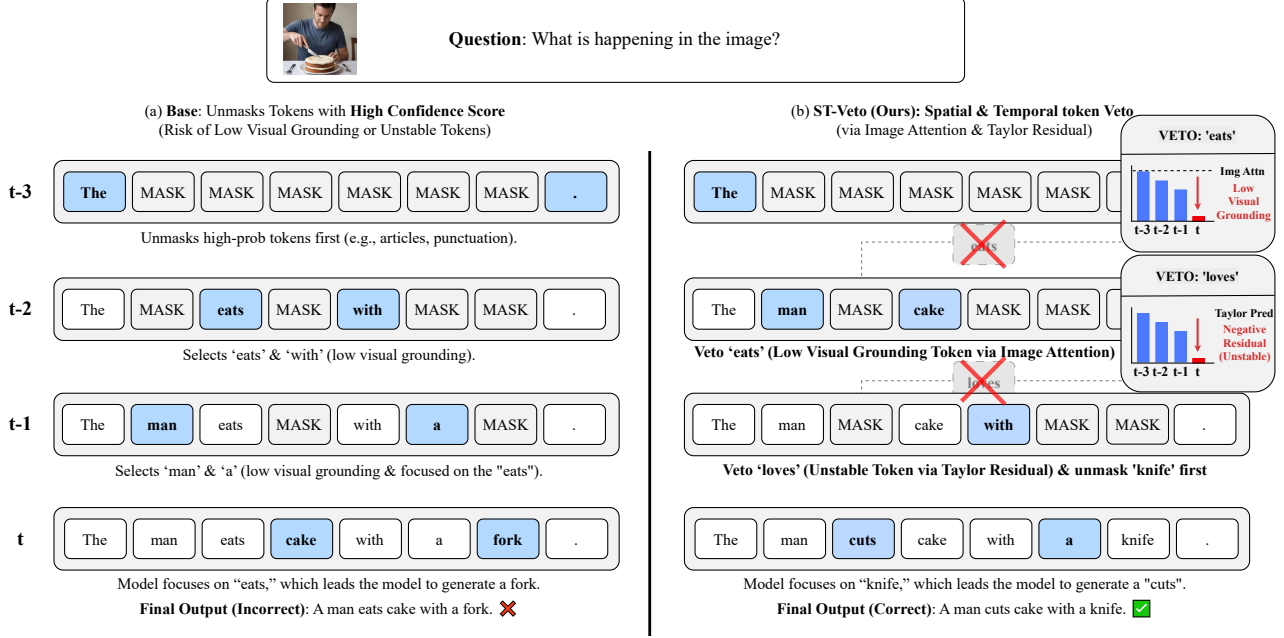


Figure 1. Overview of our work. The base decoder unmasks tokens according to current-step confidence, whereas ST-Veto applies temporal-stability and visual-grounding checks before committing tokens to the denoising trajectory.

2022) while allowing sufficiently long generation lengths. Crucially, we observe that directly applying state-of-the-art (SOTA) reasoning methods developed for VLMs is ineffective under the diffusion generation paradigm (Mitra et al., 2024; Zheng et al., 2023; Zhang et al., 2023; Xu et al., 2025). These methods assume an autoregressive chain in which intermediate rationales become stable conditioning context for subsequent tokens. In contrast, dMLLMs repeatedly revise many token positions in parallel, so a token selected at one step can become an unreliable anchor if it is chosen only because it appears confident under the current partially denoised context. These observations motivate a new approach tailored to dMLLMs rather than adapting AR-style reasoning methods.

Our approach leverages a unique characteristic of dMLLMs distinct from AR models: the ability to observe the evolution of all token positions across diffusion time steps. Unlike the generation of AR models, dMLLMs observe all masked positions at every step, unmasking confident tokens while remasking the rest for the subsequent iteration (Sahoo et al., 2024b; Austin et al., 2021). This global visibility suggests that reliable reasoning should not depend only on the instantaneous confidence of a token. Instead, a token should be committed when it is stable across denoising time and meaningfully grounded in the visual input. Capitalizing on this mechanism, we propose **Spatio-Temporal Token Veto (ST-Veto)**, a training-free decoding-time method that exploits spatio-temporal token information to guide generation.

ST-Veto operates via two key mechanisms. First, it tracks

token confidence values across diffusion steps and predicts the confidence at the next step using a second-order Taylor expansion. By comparing the actual confidence with the predicted value, ST-Veto identifies unstable tokens whose confidence deviates from the expected trajectory and vetoes them during unmasking selection. Second, it employs image attention mass to veto tokens with low visual grounding, preventing tokens from being unmasked solely due to high textual confidence without sufficient visual support (Favero et al., 2024; Kang et al., 2025; Jiang et al., 2025). Vetoed tokens are replaced only with near-boundary candidates that satisfy both temporal and visual reliability criteria. As a result, ST-Veto encourages the model to build its subsequent context from tokens that are spatio-temporally stable and semantically meaningful.

Across all evaluated models, benchmarks, and settings, ST-Veto consistently outperforms both the baseline decoding strategy and SOTA reasoning methods originally designed for VLMs. Extensive analyses further confirm that ST-Veto functions as intended, guiding dMLLMs toward generation trajectories with higher overall confidence. Notably, ST-Veto requires no additional training and introduces no additional generation cost, while achieving improvements of up to 9%. Our contributions are summarized as follows:

- We systematically evaluate the reasoning behavior of dMLLMs under long-generation settings and show that VLM-based reasoning methods do not transfer effectively to diffusion decoding.

- We propose ST-Veto, a novel training-free decoding-time method that prioritizes spatio-temporally stable tokens by combining confidence-evolution dynamics with visual grounding signals.

Conflict of Interest Disclosure. The authors declare no financial conflicts of interest related to this work.

2. Background

2.1. Diffusion Large Language Models

Diffusion Large Language Models (dLLMs) adapt the diffusion paradigm originally proposed for generating continuous data such as images or audio (Ho et al., 2020; Rombach et al., 2022) to language generation. Early studies applied diffusion to continuous text embeddings (Li et al., 2022; Gong et al., 2022); however, directly modeling discrete token distributions proved challenging, limiting performance. Subsequently, *discrete diffusion* approaches were introduced to learn the probabilistic masking-and-restoration process directly at the token level (Austin et al., 2021; Sahoo et al., 2024a), substantially improving both generation quality and training stability. Recently, dLLMs such as LLaDA (Nie et al., 2025) and Dream (Ye et al., 2025) have emerged, offering a new paradigm that balances inference speed and generation quality through parallel token restoration, effectively addressing the speed–quality trade-off.

Formally, given a discrete token sequence of length L , $X_0 = [X_0^1, X_0^2, \dots, X_0^L]$, we consider a continuous-time masking process parameterized by $t \in [0, 1]$. Let $q(X_t | X_s)$ denote the forward noising (masking) transition kernel for $1 \geq t \geq s \geq 0$, which progressively replaces a subset of tokens with a special mask token $[M]$. At $t = 1$, the sequence becomes fully masked, i.e., $X_1 = [M, M, \dots, M]$. A model p_θ parameterizes the reverse denoising process, learning to reconstruct less-masked states from more-masked ones (e.g., $p_\theta(X_s | X_t)$ for $s < t$) (Sahoo et al., 2024a; Nie et al., 2025).

During inference, sampling starts from the fully masked sequence X_1 and iteratively transitions to less-masked states by predicting masked tokens using the learned reverse model. Concretely, given the current state X_t , the model predicts all masked positions in parallel via $p_\theta(\cdot | X_t)$. Then, following a masking schedule, some tokens are *frozen* (kept) while the remaining tokens are *remasked* for further refinement. One simple example is to remask a fraction determined by a schedule $\gamma(t \rightarrow s)$ (e.g., a function of t and s) while keeping the others fixed (Sahoo et al., 2024a). Repeating this procedure gradually reduces the masking level and ultimately yields a complete token sequence.

2.2. Diffusion Multimodal Large Language Models

Diffusion MLLMs extend discrete diffusion language modeling to conditional generation with multimodal inputs, such as an image I and a prompt P . Models like LaViDa (Li et al., 2025) and MMaDA (Yang et al., 2025) learn reverse denoising as a masked-token prediction task conditioned on multimodal signals.

LaViDa encodes an image I into continuous visual features and trains a diffusion language model to predict the original text sequence X_0 from a partially masked X_t given I and P . The training objective is defined as (Li et al., 2025):

$$\mathcal{L}_{\text{LaViDa}} = -\mathbb{E}_{t,I,P,X_0,X_t} \left[w(t) \log p_\theta(X_0 | I, P, X_t) \right], \quad (1)$$

where $w(t)$ is a time-dependent reweighting (often chosen as $w(t) = 1/t$). With masked positions $\mathcal{M}_t = \{i | X_t^i = [M]\}$, the loss effectively applies only to those positions. To fit the constraints of the model, we reformulate the objective as:

$$\mathcal{L}_{\text{LaViDa}} = -\mathbb{E}_{t,I,P,X_0,X_t} \left[w(t) \sum_{i \in \mathcal{M}_t} \log p_\theta(X_0^i | I, P, X_t) \right]. \quad (2)$$

In contrast, MMaDA unifies text and image into a single discrete token sequence (Yang et al., 2025). With discrete image tokens $V_0 = [V_0^1, \dots, V_0^{L_v}]$ and text tokens $X_0 = [X_0^1, \dots, X_0^{L_x}]$, the unified representation is:

$$U_0 = [P; X_0; V_0]. \quad (3)$$

The forward process masks a subset of positions to form U_t , and the model predicts the masked tokens. The unified pretraining objective is given by:

$$\mathcal{L}_{\text{MMaDA}} = -\mathbb{E}_{t,U_0,U_t} \left[\frac{1}{|\mathcal{M}_t|} \sum_{i \in \mathcal{M}_t} \log p_\theta(U_0^i | U_t) \right], \quad (4)$$

where $t \sim \mathcal{U}(0, 1)$.

Unlike LaViDa, which uses continuous visual features for conditioning, MMaDA tokenizes images discretely and optimizes a single diffusion objective over both text and image tokens, enabling unified multimodal understanding and text-to-image generation.

2.3. Reasoning in dMLLMs

Current research on multimodal reasoning primarily targets autoregressive (AR) models, employing Chain-of-Thought (CoT) strategies that rely on generating intermediate rationales sequentially (Wei et al., 2022; Zhang et al., 2023).

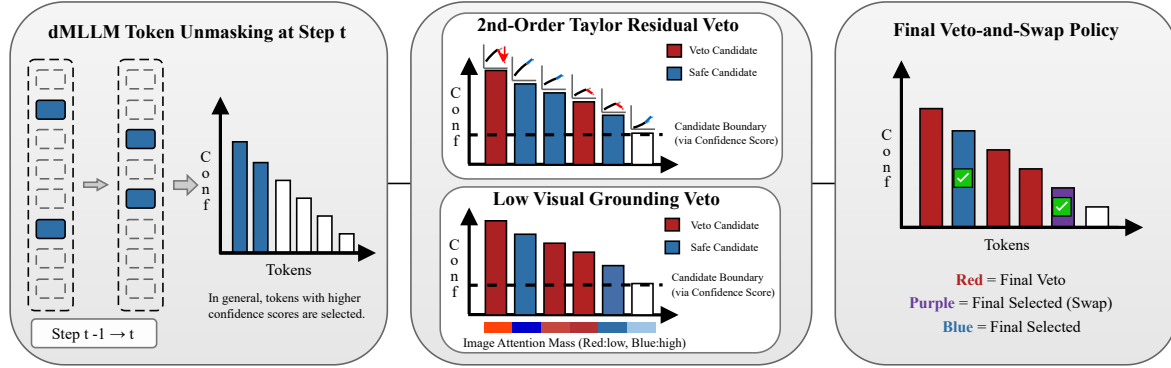


Figure 2. Overview of ST-Veto. Tokens are vetoed based on the Taylor residual and low visual grounding, and are replaced by safe tokens that are unmasked instead.

However, these methods are structurally incompatible with the diffusion generation paradigm. Unlike AR models that condition current generation on a fixed history of preceding tokens, dMLLMs update all tokens simultaneously through iterative denoising (Sahoo et al., 2024b). Consequently, the standard “generate-then-condition” mechanism fails in diffusion decoding because there is no stable, causal sequence to anchor intermediate reasoning steps during the bidirectional refinement process. This fundamental mismatch underscores the necessity for diffusion-native reasoning approaches that leverage the global visibility of token evolution, rather than forcing sequential dependencies.

3. Method

We operationalize the self-correction capability of diffusion models through a mechanism we term “Veto-and-Swap”. A common diffusion decoding policy unmask the currently most confident tokens at each step. However, an unmasking decision made only from the instantaneous context at step t can be unreliable: a token may be highly ranked because it is locally obvious in the current partially denoised sequence, while its confidence trajectory is unstable or its prediction is weakly supported by the image. ST-Veto therefore seeks to secure tokens that are both temporally stable and spatially grounded before they become fixed context for later denoising steps. Our approach is motivated by three key insights derived from the properties of diffusion models:

- **Stability via Veto-and-Swap:** Directly manipulating logits or attention distributions can destabilize the diffusion process. Instead, a discrete veto-and-swap policy offers a more robust control mechanism.
- **Temporal Stability:** Unlike autoregressive models, dMLLMs provide visibility into all masked token positions at every step t , allowing us to leverage the evolution of confidence over time rather than relying only

on the current-step score.

- **Visual Anchoring:** Since dMLLMs generate tokens in parallel, we can prioritize unmasking tokens that are supported by visual evidence, using them as spatial anchors to guide subsequent generation.

Notation. Let $\mathbf{c}^{(t)} \in \mathbb{R}^n$ denote the confidence score vector of the currently masked token positions at diffusion step t , where n is the number of masked positions. Each element $c_i^{(t)}$ is the model confidence assigned to the predicted token at position i . We aim to unmask k positions at this step. Let $\mathcal{I}$ be the set of indices corresponding to image tokens, and let $A^{(\ell)} \in \mathbb{R}^{L_{\text{seq}} \times L_{\text{seq}}}$ denote the head-averaged attention matrix at Transformer layer ℓ , where L_{seq} is the total multimodal sequence length. We denote the number of Transformer layers by N_{lay} .

3.1. Taylor Residual (Temporal Instability Signal)

In diffusion-based generation, token-level confidence scores are expected to evolve smoothly as the sequence is progressively denoised. However, abrupt fluctuations often signal instability: a token may look confident under the current noisy context but become unsupported once neighboring tokens are refined. Such premature commitments can produce unstable anchors for later steps. To identify these cases, we predict the expected confidence via a second-order Taylor expansion using finite-difference velocity and acceleration computed from a local 4-step history over the same masked positions, $\{\mathbf{c}^{(t-3)}, \dots, \mathbf{c}^{(t)}\}$:

$$\hat{\mathbf{c}}^{(t)} = \mathbf{c}^{(t-1)} + (\mathbf{c}^{(t-1)} - \mathbf{c}^{(t-2)}) + \frac{1}{2} [(\mathbf{c}^{(t-1)} - \mathbf{c}^{(t-2)}) - (\mathbf{c}^{(t-2)} - \mathbf{c}^{(t-3)})]. \quad (5)$$

The instability signal (Taylor residual) for token i is then defined as the deviation from this trajectory:

$$e_i = c_i^{(t)} - \hat{c}_i^{(t)}. \quad (6)$$

Here, large negative residuals ($e_i \ll 0$) indicate that the confidence has dropped significantly below expectation, signaling temporal instability.

3.2. Image Attention (Grounding Signal)

Temporal stability alone does not ensure that an unmasked token is semantically grounded in the visual input. A token can be stable because it is a frequent or textually plausible continuation, while still being weakly supported by the image. Unlike sequential generation, dMLLMs can commit to token positions in any order, so ST-Veto can prefer tokens that are meaningful in the current multimodal context and visually supported before they influence later denoising. We measure visual grounding by aggregating the attention mass assigned to image tokens $\mathcal{I}$ across a set of intermediate layers:

$$\mathcal{L}_{\text{attn}} = \left\{ \frac{N_{\text{lay}}}{3}, \frac{N_{\text{lay}}}{2}, \frac{2N_{\text{lay}}}{3} \right\}. \quad (7)$$

When these indices are non-integers, we use the nearest valid layer index. For a token i at layer ℓ , we define the image attention ratio $\rho_i^{(\ell)}$ as:

$$\rho_i^{(\ell)} = \frac{\sum_{j \in \mathcal{I}} A_{i,j}^{(\ell)}}{\sum_{p=1}^{L_{\text{seq}}} A_{i,p}^{(\ell)}}. \quad (8)$$

The final grounding signal m_i is obtained by averaging across the selected layers:

$$m_i = \frac{1}{|\mathcal{L}_{\text{attn}}|} \sum_{\ell \in \mathcal{L}_{\text{attn}}} \rho_i^{(\ell)}. \quad (9)$$

3.3. Veto-and-Swap Policy

A standard decoding strategy is to unmask the set of tokens with the highest confidence scores at the current step. This greedy rule is efficient, but it only reflects the current partially denoised context. ST-Veto instead treats the top- k set as a proposal and checks whether each proposed token is spatio-temporally reliable. Let $\mathcal{T}_k$ denote the indices of the top- k tokens:

$$\mathcal{T}_k = \left\{ i \mid \text{rank}_{\downarrow}(c_i^{(t)}) \leq k \right\}, \quad (10)$$

where $\text{rank}_{\downarrow}$ ranks confidence scores in descending order. Let $c_{(k)} = \min_{i \in \mathcal{T}_k} c_i^{(t)}$ represent the confidence threshold. While efficient, this greedy selection often includes unstable or weakly grounded tokens. We therefore apply a conservative policy based on robust statistics.

Robust statistics. For a signal $X \in \{e, m\}$, we compute robust location and scale estimates: $\mu_X = \text{median}(X)$ and $\sigma_X = \text{MAD}(X)$.

Veto and Safe Criteria. A token i is flagged for veto if either its temporal residual or visual grounding score deviates negatively from the population distribution:

$$\mathcal{C}_{\text{veto}}(i) \iff (e_i < \mu_e - \sigma_e) \vee (m_i < \mu_m - \sigma_m). \quad (11)$$

Conversely, a token is considered safe only if it satisfies both stricter temporal stability and visual grounding requirements:

$$\mathcal{C}_{\text{safe}}(i) \iff (e_i \geq \mu_e + \sigma_e) \wedge (m_i \geq \mu_m + \sigma_m). \quad (12)$$

Veto and Swap sets. We define the veto set $\mathcal{V}$ as the risky tokens currently within the top- k selection:

$$\mathcal{V} = \{i \in \mathcal{T}_k \mid \mathcal{C}_{\text{veto}}(i)\}. \quad (13)$$

To fill the vacated slots, we identify a set of swap candidates $\mathcal{S}$. To prevent destabilization from low-confidence tokens, candidates must be safe and reside within a ‘‘candidate line’’ (a near-boundary region around $c_{(k)}$):

$$\mathcal{S} = \left\{ i \notin \mathcal{T}_k \mid \mathcal{C}_{\text{safe}}(i) \wedge |c_i^{(t)} - c_{(k)}| \leq \sigma_c \right\}, \quad (14)$$

where $\sigma_c = \text{MAD}(c^{(t)})$.

Final Selection. We replace the tokens in $\mathcal{V}$ with the highest-confidence candidates from $\mathcal{S}$. If $|\mathcal{S}| < |\mathcal{V}|$, we only perform partial swaps. Thus, ST-Veto does not force a replacement when no sufficiently stable and grounded alternative exists. The final mask is updated accordingly for the next step, allowing the model to build subsequent context from tokens that are more reliable across time and more meaningful with respect to the image.

4. Experimental Setup

Benchmarks. We evaluate our proposed method on four diverse benchmarks: **M3CoT** (Chen et al., 2024), **ScienceQA** (Lu et al., 2022), **MMBench** (Liu et al., 2024), and **V*Bench** (Wu & Xie, 2023). **M3CoT** is designed to assess multimodal chain-of-thought (CoT) reasoning across broad domains, including science, mathematics, and common-sense. **ScienceQA** evaluates knowledge-based multimodal reasoning by integrating scientific concepts with visual information. **MMBench** comprises diverse evaluation splits to measure general visual perception and visual reasoning capabilities. **V*Bench** focuses on high-resolution visual question answering (VQA), emphasizing fine-grained visual understanding. We follow the official evaluation protocols and report results on the standard splits for all benchmarks.

Table 1. Main results table. X/Y denotes the generation length L and the number of steps T . Bold indicates the best performance. All experiments were conducted three times with different random seeds, and the reported results are averaged.

Model	Method	M ³ CoT		MMBench		SQA-IMG		V* Bench	
		128/64	256/128	128/64	256/128	128/64	256/128	128/64	256/128
LaViDa	Base-Confidence	27.96	29.55	33.96	31.55	45.81	40.06	35.60	37.17
	Base-Margin	27.88	28.22	32.99	31.68	46.01	38.99	35.08	37.70
	Base-Entropy	27.91	29.23	32.67	31.92	45.89	39.17	34.55	36.65
	DDCoT	26.99	29.11	33.02	30.98	45.99	38.81	35.08	35.60
	CCoT	26.57	28.98	33.19	31.04	45.21	38.90	36.65	36.65
	SCAFFOLD	26.61	28.81	33.24	31.22	44.98	39.25	35.60	35.08
	ICoT	27.82	29.29	33.22	31.43	45.68	39.17	37.17	37.17
	ST-Veto (Ours)	30.21	31.87	38.05	38.25	47.80	48.54	42.94	41.36
	Base-Confidence	26.32	25.11	33.75	32.04	44.72	41.70	34.03	31.41
	Base-Margin	26.22	25.54	33.22	31.25	44.99	41.55	32.98	32.46
MMaDA	Base-Entropy	26.36	25.32	33.53	31.87	44.23	41.98	33.51	31.41
	DDCoT	25.44	24.45	33.21	30.04	43.99	39.98	33.51	29.84
	CCoT	26.10	24.88	34.01	31.55	44.81	40.12	34.55	29.32
	SCAFFOLD	25.88	24.91	33.88	30.98	44.92	40.33	31.41	28.80
	ICoT	26.01	24.78	33.93	32.44	44.52	40.47	34.55	31.41
	ST-Veto (Ours)	27.52	27.74	36.55	36.74	47.25	45.31	41.36	37.17
	Base-Confidence	26.32	25.11	33.75	32.04	44.72	41.70	34.03	31.41

Table 2. Time cost table. The unit is inference per second, and the values are averaged over runs on MMBench.

Method	MMaDA		LaViDa	
	128/64	256/128	128/64	256/128
Base-Confidence	5.68	11.32	4.87	12.88
Base-Margin	5.65	11.28	4.85	12.84
Base-Entropy	5.68	11.33	4.88	12.89
DDCoT	10.87	19.75	9.89	22.95
CCoT	10.93	19.48	9.92	22.94
SCAFFOLD	10.99	19.43	9.85	22.91
ICoT	10.96	19.89	9.87	22.87
ST-Veto	5.73	11.39	4.89	12.92

Models and Baselines. Our experiments are conducted on two diffusion-based multimodal large language models (dMLLMs) exhibiting strong reasoning capabilities: **LaViDa-llada-reason** (Li et al., 2025) and **MMaDA-8B-MixCoT** (Yang et al., 2025). Both models are fine-tuned to generate rationales and produce progressive outputs via an unmasking-based generation process. For rigorous comparisons, we establish two groups of baselines. First, we compare decoding-policy baselines using three remasking strategies: **Low-confidence (Low-conf)**, **Entropy**, and **Margin** (Sahoo et al., 2024a). Second, we include CoT-style prompting baselines that have demonstrated strong performance in autoregressive VLMs: **CCoT** (Mittra et al., 2024) and **DDCoT** (Zheng et al., 2023), as well as more recent approaches that explicitly enhance vision-language coordination, including **ICoT** (Gao et al., 2025) and **SCAFFOLD** (Lei et al., 2024). For all prompting baselines, we apply their original prompting templates (and visual prompt-

ing schemes when applicable) while employing Low-conf remasking as the decoding policy for fair comparison.

Implementation Details. We evaluate our method by analyzing the speed-quality trade-off—a key advantage of dMLLMs—across varying target output lengths and denoising timesteps. Specifically, we fix the ratio L/T to a standard setting of 0.5, where L denotes the target generation length and T represents the number of denoising iterations. To ensure strict reproducibility, we perform deterministic decoding by selecting the argmax token at each denoising step (i.e., no temperature scaling). Unless otherwise specified, Low-conf (Sahoo et al., 2024a) serves as the default remasking policy. Additional prompt configurations and detailed experimental settings are provided in Appendix A.1.

5. Results

5.1. Main Results

Table 1 presents the main evaluation results across four benchmarks. We report the average performance over three random seeds. Experimental settings are denoted as L/T , where L represents the target generation length and T indicates the number of denoising steps. Unless otherwise specified, “think mode” refers to enabling rationale generation during decoding, with the maximum number of new tokens capped by L .

Performance of Baselines. As detailed in Table 1, both LaViDa and MMaDA struggle to attain competitive performance when “think mode” is enabled with $L \in \{128, 256\}$. Notably, they exhibit limited performance even on MM-

Table 3. Ablation study table. X/Y denotes the generation length L and the number of steps T . Bold indicates the best performance. *w/o Taylor* and *w/o Img* apply the same Veto-and-Swap policy.

Model	Setting	M ³ CoT		MMBench		SQA-IMG		V* Bench	
		128/64	256/128	128/64	256/128	128/64	256/128	128/64	256/128
LaViDa	Base	27.96	29.55	33.96	31.55	45.81	40.06	35.60	37.17
	w/o Taylor	28.77	29.64	36.04	35.94	45.86	46.52	38.22	39.27
	w/o Img	29.68	29.11	36.22	35.22	47.47	47.89	41.36	38.22
	ST-Veto	30.21	31.87	38.05	38.25	47.80	48.54	42.94	41.36
MMaDA	Base	26.32	25.11	33.75	32.04	44.72	41.70	34.03	31.41
	w/o Taylor	26.66	25.99	35.12	35.52	47.15	43.99	39.79	32.84
	w/o Img	26.88	26.24	34.98	35.10	45.56	43.87	37.17	33.51
	ST-Veto	27.52	27.74	35.55	35.74	47.25	45.31	41.36	37.17

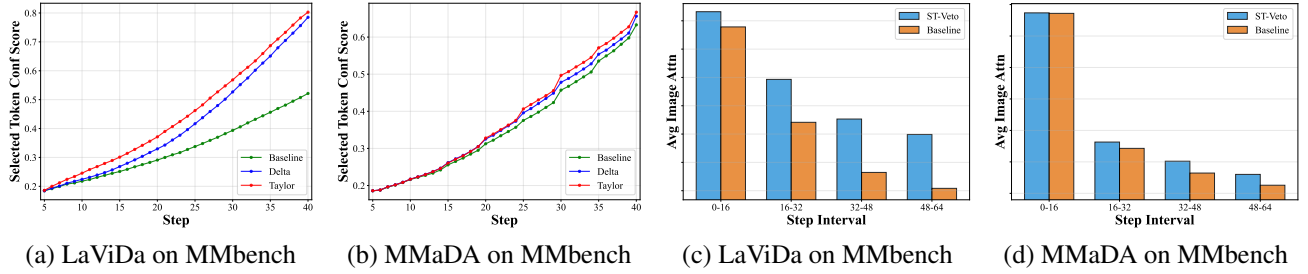


Figure 3. (a) and (b) show the confidence scores of tokens selected at each step for different methods. (c) and (d) present the relative values of the Average Image Attention.

Table 4. Sensitivity to history length on LaViDa. $H = 4$ provides the best cost-performance trade-off; $H = 5$ occasionally matches or exceeds it but requires a higher-order finite-difference estimate.

Dataset	L/T	Base	Delta	1st	2nd	3rd
MMBench	128/64	33.96	36.98	37.38	38.05	38.11
	256/128	31.55	36.89	37.98	38.25	38.12
SQA-IMG	128/64	45.81	46.69	47.25	47.80	47.91
	256/128	40.06	46.36	47.60	48.54	47.55
M ³ CoT	128/64	27.96	28.69	29.21	30.21	30.18
	256/128	29.55	30.03	30.50	31.87	31.55
V* Bench	128/64	35.60	37.17	41.36	42.94	39.27
	256/128	37.17	39.27	39.27	41.36	41.36

Bench, a widely adopted benchmark for general VLM evaluation. This underscores a critical bottleneck in current dMLLMs: despite instruction tuning for reasoning (e.g., LaViDa-reason and MMaDA-MixCoT), these models often fail to maintain coherent multi-step rationales under the parallel decoding paradigm. Furthermore, standard remasking strategies (Low-conf, Entropy, Margin) show only marginal performance variance in Table 1, indicating that simple uncertainty-based heuristics are insufficient for complex multimodal reasoning.

Failure of AR-based Methods. Methods originally designed for autoregressive (AR) VLMs—including DDCoT, CCoT, and ICoT—prove ineffective when directly applied

to dMLLMs (Table 1). These approaches rely on sequential dependency, generating intermediate contexts (e.g., captions, scene graphs, or rationales) before generating the final answer. However, dMLLMs generate tokens in parallel via iterative refinement, which does not naturally support reusing intermediate tokens as fixed conditioning signals. Consequently, the “generate-then-condition” paradigm of AR methods does not transfer well to diffusion decoding.

Superiority of ST-Veto. Across all models, benchmarks, and L/T settings, ST-Veto consistently achieves the best performance (Table 1). Unlike baselines hampered by such structural mismatches, ST-Veto delivers substantial and consistent gains without requiring additional training, multi-stage generation, or direct modification of attention maps or logits. These results support our hypothesis that effective reasoning control in dMLLMs necessitates mechanisms tailored to the diffusion process rather than direct transplantation of AR-based techniques.

5.2. Efficiency

Table 2 compares the inference throughput (samples per second) of different methods on MMBench. All methods are evaluated on identical hardware with the same caching mechanisms in a training-free setting.

Computational Overhead. ST-Veto introduces negligible overhead compared to the base model. The computations

Table 5. Layer ablation for the image-attention grounding signal on LaViDa. Combining intermediate depths is more robust than using a single layer.

Dataset	L/T	Base	$L/3$	$L/2$	$2L/3$	All
MMBench	128/64	33.96	35.83	37.12	37.68	38.05
	256/128	31.55	34.21	36.08	35.74	38.25
SQA-IMG	128/64	45.81	46.95	47.40	46.85	47.80
	256/128	40.06	44.52	46.71	47.85	48.54
M ³ CoT	128/64	27.96	29.12	29.85	30.37	30.21
	256/128	29.55	30.67	31.23	30.85	31.87
V* Bench	128/64	35.60	38.74	40.31	41.36	42.94
	256/128	37.17	39.27	40.31	41.88	41.36

for second-order Taylor scoring and visual grounding are lightweight and seamlessly integrated into the decoding loop. **Comparison with AR Methods.** In contrast, adapting AR-based methods to dMLLMs typically incurs significant computational costs. To emulate their intended pipeline (context $\rightarrow$ answer), we perform two separate inference calls: one to generate the intermediate context and another to generate the final response conditioned on that context. Single-pass attempts cannot reliably reproduce the intended behavior due to the non-sequential nature of diffusion decoding. As a result, AR-based baselines effectively double the generation cost, whereas ST-Veto operates efficiently within a single diffusion inference process.

5.3. Ablation Study

To quantify the contribution of each component, we conduct an ablation study on the Second-Order Taylor Veto and the Visual Grounding Veto (Table 3). Here, w/o Taylor disables the Taylor Veto (using only Visual Grounding), and w/o Img disables the Visual Grounding Veto (using only Taylor); both variants retain the Veto-and-Swap policy.

We observe that employing either component individually yields improvements over the base model on most benchmarks, indicating that both Hessian-based confidence (Taylor) and visual consistency (Visual Grounding) are beneficial for dMLLM reasoning. Crucially, the full ST-Veto achieves the highest performance, demonstrating complementary effects from enforcing both textual logicity and visual grounding. The modular design further facilitates plug-and-play integration with other decoding strategies.

Table 4 further examines the temporal order used by the Taylor-based veto on LaViDa. All history-aware variants outperform the baseline, indicating that temporal confidence dynamics are broadly useful for dMLLM decoding. Among them, the proposed second-order setting ($H = 4$) achieves the strongest overall trade-off: it is best in most settings and avoids the additional higher-order finite-difference estimate required by $H = 5$.

Table 6. Sensitivity to the robust threshold on LaViDa. The default MAD cutoff is consistently strong across benchmarks and generation settings.

Dataset	L/T	Base	0.5MAD	MAD	1.5MAD
MMBench	128/64	33.96	36.62	38.05	37.24
	256/128	31.55	36.45	38.25	36.12
SQA-IMG	128/64	45.81	47.12	47.80	47.80
	256/128	40.06	47.45	48.54	47.22
M ³ CoT	128/64	27.96	29.96	30.21	28.88
	256/128	29.55	31.22	31.87	31.22
V* Bench	128/64	35.60	39.27	42.94	41.36
	256/128	37.17	40.61	41.36	38.22

Table 5 analyzes which layers should be used for the visual grounding signal. Single-layer variants already improve over the baseline in most settings, but no individual depth is uniformly optimal across benchmarks. Averaging the intermediate layers ($L/3$, $L/2$, and $2L/3$) gives the most reliable overall behavior, suggesting that different depths provide complementary visual evidence for selecting spatially meaningful tokens.

Table 6 evaluates the robustness of the MAD-based thresholds used in the veto and safe criteria. The relaxed 0.5MAD setting allows more tokens to enter the veto-and-swap procedure, while 1.5MAD is more conservative and can prevent useful swaps. The default MAD cutoff achieves the strongest or tied-strongest performance in nearly all settings, supporting a distribution-adaptive threshold that balances correction and stability.

6. Analysis

6.1. Methodological Analysis

Table 7 reports, for each model and benchmark, the proportions of tokens in the top- k set that are actually selected through the veto-and-swap procedure, as well as the veto and safe ratios among candidate tokens that meet a certain confidence threshold. Specifically, when unmasking k tokens at each step, the top- k column shows the fraction of tokens that are vetoed and subsequently replaced by candidate tokens. Both LaViDa and MMaDA exhibit similar trends across all benchmarks. Although the veto-and-swap criteria can be adjusted to increase the number of swapped tokens, excessively overriding the model’s inherent confidence estimates is undesirable, as assumed in our method design.

The veto ratio among candidate tokens is substantially higher than that in the top- k set. Our analysis reveals a strong correlation between the proposed stability metrics and model confidence: tokens flagged by Taylor residuals or low image attention consistently exhibit lower confidence

Table 7. Analysis Table. Metrics are reported for *top-k* (Veto & Swap) and *candidate* (Veto / Safe; in percent with *Veto+Safe=100*) under each configuration (128/64 and 256/128). Candidate ratios for each model/setting are adjusted to be close to the *V* Bench* ratio.

Model	Benchmark	128/64			256/128		
		top-k		candidate	top-k		candidate
		Veto & Swap	Veto		Veto & Swap	Veto	
<i>LaViDa</i>	M ³ CoT	4.8±1.6	18.2±5.2	81.8±5.2	2.6±0.9	19.3±3.9	80.7±3.9
	MMBench	5.3±2.2	19.0±6.4	81.0±6.4	3.1±1.3	20.1±3.2	79.9±3.2
	SQA-IMG	5.1±2.1	18.7±5.6	81.3±5.6	2.9±1.2	19.8±3.1	80.2±3.1
	V* Bench	4.8±6.1	18.4±5.9	81.6±5.9	2.2±3.0	19.5±2.9	80.5±2.9
<i>MMaDA</i>	M ³ CoT	5.8±1.3	14.5±3.0	85.5±3.0	7.4±1.8	11.8±3.5	88.2±3.5
	MMBench	5.2±1.0	14.2±2.4	85.8±2.4	6.9±1.5	11.5±2.7	88.5±2.7
	SQA-IMG	3.9±0.8	13.6±2.3	86.4±2.3	5.5±1.2	10.9±3.1	89.1±3.1
	V* Bench	6.1±2.9	14.7±2.0	85.3±2.0	6.3±3.7	11.2±2.8	88.8±2.8

scores. Consequently, the candidate set naturally yields a higher veto ratio, as it contains tokens with lower confidence scores than those in the *top-k* set. When comparing LaViDa and MMaDA, MMaDA shows a higher proportion of safe tokens. This aligns with the larger performance gains observed for LaViDa relative to MMaDA in Figure 3 and is likely attributable to differences in their training architectures, particularly in how visual information is processed.

6.2. Temporal Dynamics Analysis

The core idea of our method is that token unmasking decisions at step t should not be made solely based on the instantaneous state at that step, but should instead account for how token confidence evolves over time as diffusion progresses. To demonstrate the effectiveness of this insight, Figure 3 compares the confidence scores of tokens selected at each step by methods that incorporate temporal information with those that do not.

The results show that methods using Taylor-based veto not only outperform the baseline, but even a simple *delta veto*, which considers only the confidence difference between the current and the immediately preceding step, achieves higher confidence scores than the baseline. The delta signal serves as a first-order alternative to the Taylor residual. These findings indicate that vetoing temporally unstable tokens and swapping them with more stable ones guides the generation toward output paths with higher overall confidence. This trend is consistently observed in all models, with a stronger effect in LaViDa. Moreover, a veto-and-swap policy based solely on confidence at step $t - 1$ is less effective than one that leverages second-order Taylor predictions, highlighting the benefit of modeling higher-order temporal dynamics.

6.3. Spatial Dynamics Analysis

Figure 3 presents the results of measuring the Average Image Attention Ratio for the baseline and ST-Veto. Since outliers may appear at individual steps, we visualize the

results by grouping steps into intervals to better capture the overall trend. The results show that ST-Veto consistently achieves higher image attention ratios than the baseline across all steps. This can be interpreted as the effect of vetoing certain tokens that, despite having high confidence scores, exhibit weak visual grounding.

As the model progressively generates and conditions on more text tokens, the overall average image attention naturally decreases as diffusion steps accumulate. In contrast, the gap between the baseline and ST-Veto widens over time, a pattern that closely mirrors the confidence score dynamics observed in the temporal analysis. These findings suggest that the veto-and-swap policy guides the model toward generation paths that are superior not only in terms of confidence scores but also in visual grounding.

7. Conclusion

In this work, we investigated how to improve the reasoning capabilities of dMLLMs. Through extensive empirical analysis, we showed that reasoning methods designed for autoregressive models—especially those relying on sequential token dependencies—do not transfer well to diffusion decoding due to fundamental differences in the generation process. To address this gap, we proposed ST-Veto, a training-free decoding-time strategy tailored to dMLLMs. ST-Veto leverages the evolution of token confidence across denoising steps and visual grounding signals to veto unstable or weakly grounded tokens during unmasking, thereby prioritizing more stable and visually supported generations. Experiments across models and benchmarks demonstrate consistent improvements over standard diffusion decoding and adapted VLM reasoning baselines, achieving gains of up to 9% without additional model training or extra denoising steps. We believe these findings establish a solid foundation for diffusion-native reasoning and encourage future research toward more reliable multimodal inference.

Impact Statement

This paper proposes a training-free inference-time method to improve reasoning in diffusion-based multimodal large language models (dMLLMs) by refining token unmasking decisions. Our contribution is methodological: we introduce a veto-and-swap policy that uses temporal confidence dynamics and image-attention-based grounding signals to reject unreliable tokens and replace them with safer candidates, improving inference stability without updating model parameters.

The method can enable more accurate and better-grounded multimodal outputs without additional training or generation cost. However, it does not eliminate common risks of multimodal generative systems, including incorrect or biased outputs and potential misuse. Because the approach relies on internal signals (confidence and attention) that are imperfect proxies for correctness, responsible deployment should include standard evaluation and safeguards, with human oversight in high-stakes settings.

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References

- Alayrac, J.-B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., et al. Flamingo: a visual language model for few-shot learning. *Advances in neural information processing systems*, 35:23716–23736, 2022.
- Austin, J., Johnson, D. D., Ho, J., Tarlow, D., and Van Den Berg, R. Structured denoising diffusion models in discrete state-spaces. *Advances in neural information processing systems*, 34:17981–17993, 2021.
- Chen, Q., Qin, L., Zhang, J., Chen, Z., Xu, X., and Che, W. M³ cot: A novel benchmark for multi-domain multi-step multi-modal chain-of-thought. *arXiv preprint arXiv:2405.16473*, 2024.
- Favero, A., Zancato, L., Trager, M., Choudhary, S., Perera, P., Achille, A., Swaminathan, A., and Soatto, S. Multimodal hallucination control by visual information grounding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 14303–14312, 2024.
- Fu, C., Chen, P., Shen, Y., Qin, Y., Zhang, M., Lin, X., Yang, J., Zheng, X., Li, K., Sun, X., and Ji, R. MME: A comprehensive evaluation benchmark for multimodal large language models. *arXiv preprint arXiv:2306.13394*, 2023.
- Gao, J., Li, Y., Cao, Z., and Li, W. Interleaved-modal chain-of-thought. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 19520–19529, 2025.
- Gong, S., Li, M., Feng, J., Wu, Z., and Kong, L. Diffuseq: Sequence to sequence text generation with diffusion models. *arXiv preprint arXiv:2210.08933*, 2022.
- Ho, J., Jain, A., and Abbeel, P. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.
- Huang, Q., Dong, X., Zhang, P., Wang, B., He, C., Wang, J., Lin, D., Zhang, W., and Yu, N. OPERA: Alleviating hallucination in multi-modal large language models via over-trust penalty and retrospection-allocation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- Ji, Z., Yu, T., Xu, Y., Lee, N., Ishii, E., and Fung, P. Towards mitigating llm hallucination via self reflection. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 1827–1843, 2023.
- Jiang, Z., Chen, J., Zhu, B., Luo, T., Shen, Y., and Yang, X. Devils in middle layers of large vision-language models: Interpreting, detecting and mitigating object hallucinations via attention lens. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 25004–25014, 2025.
- Kang, S., Kim, J., Kim, J., and Hwang, S. J. See what you are told: Visual attention sink in large multimodal models. *arXiv preprint arXiv:2503.03321*, 2025.
- Langley, P. Crafting papers on machine learning. In Langley, P. (ed.), *Proceedings of the 17th International Conference on Machine Learning (ICML 2000)*, pp. 1207–1216, Stanford, CA, 2000. Morgan Kaufmann.
- Lei, X., Yang, Z., Chen, X., Li, P., and Liu, Y. Scaffolding coordinates to promote vision-language coordination in large multi-modal models. *arXiv preprint arXiv:2402.12058*, 2024.
- Li, C., Wong, C., Zhang, S., Usuyama, N., Liu, H., Yang, J., Naumann, T., Poon, H., and Gao, J. Llava-med: Training a large language-and-vision assistant for biomedicine in one day. *Advances in Neural Information Processing Systems*, 36:28541–28564, 2023a.
- Li, J., Li, D., Savarese, S., and Hoi, S. Blip-2: Bootstrapping language-image pre-training with frozen image encoders

- and large language models. In *International conference on machine learning*, pp. 19730–19742. PMLR, 2023b.
- Li, S., Kallidromitis, K., Bansal, H., Gokul, A., Kato, Y., Kozuka, K., Kuen, J., Lin, Z., Chang, K.-W., and Grover, A. Lavid: A large diffusion language model for multi-modal understanding. *arXiv preprint arXiv:2505.16839*, 2025.
- Li, X., Thickstun, J., Gulrajani, I., Liang, P. S., and Hashimoto, T. B. Diffusion-lm improves controllable text generation. *Advances in neural information processing systems*, 35:4328–4343, 2022.
- Li, Y., Du, Y., Zhou, K., Wang, J., Zhao, X., and Wen, J.-R. Evaluating object hallucination in large vision-language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023c.
- Liu, Y., Duan, H., Zhang, Y., Li, B., Zhang, S., Zhao, W., Yuan, Y., Wang, J., He, C., Liu, Z., et al. Mmbench: Is your multi-modal model an all-around player? In *European conference on computer vision*, pp. 216–233. Springer, 2024.
- Lu, P., Mishra, S., Xia, T., Qiu, L., Chang, K.-W., Zhu, S.-C., Tafjord, O., Clark, P., and Kalyan, A. Learn to explain: Multimodal reasoning via thought chains for science question answering. *Advances in Neural Information Processing Systems*, 35:2507–2521, 2022.
- Mitra, C., Huang, B., Darrell, T., and Herzig, R. Compositional chain-of-thought prompting for large multimodal models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 14420–14431, 2024.
- Nie, S., Zhu, F., You, Z., Zhang, X., Ou, J., Hu, J., Zhou, J., Lin, Y., Wen, J.-R., and Li, C. Large language diffusion models. *arXiv preprint arXiv:2502.09992*, 2025.
- Pan, L., Saxon, M., Xu, W., Nathani, D., Wang, X., and Wang, W. Y. Automatically correcting large language models: Surveying the landscape of diverse self-correction strategies. *arXiv preprint arXiv:2308.03188*, 2023.
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., and Ommer, B. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 10684–10695, 2022.
- Sahoo, S., Arriola, M., Schiff, Y., Gokaslan, A., Marroquin, E., Chiu, J., Rush, A., and Kuleshov, V. Simple and effective masked diffusion language models. *Advances in Neural Information Processing Systems*, 37:130136–130184, 2024a.
- Sahoo, S., Arriola, M., Schiff, Y., Gokaslan, A., Marroquin, E., Chiu, J., Rush, A., and Kuleshov, V. Simple and effective masked diffusion language models. *Advances in Neural Information Processing Systems*, 37:130136–130184, 2024b.
- Sahoo, S. S., Deschenaux, J., Gokaslan, A., Wang, G., Chiu, J., and Kuleshov, V. The diffusion duality. *arXiv preprint arXiv:2506.10892*, 2025.
- Tang, F., Liu, C., Xu, Z., Hu, M., Peng, Z., Yang, Z., Su, J., Lin, M., Peng, Y., Cheng, X., Razzak, I., and Ge, Z. Seeing far and clearly: Mitigating hallucinations in multimodal large language models with attention causal decoding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.
- Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., Chen, K., Liu, X., Wang, J., Ge, W., et al. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.
- Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., and Zhou, D. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022.
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q. V., Zhou, D., et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.
- Wu, P. and Xie, S. V*: Guided visual search as a core mechanism in multimodal llms, 2023. URL <https://arxiv.org/abs/2312.14135>.
- Xu, G., Jin, P., Wu, Z., Li, H., Song, Y., Sun, L., and Yuan, L. Llava-cot: Let vision language models reason step-by-step. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 2087–2098, 2025.
- Yang, L., Tian, Y., Li, B., Zhang, X., Shen, K., Tong, Y., and Wang, M. Mmada: Multimodal large diffusion language models. *arXiv preprint arXiv:2505.15809*, 2025.
- Ye, J., Xie, Z., Zheng, L., Gao, J., Wu, Z., Jiang, X., Li, Z., and Kong, L. Dream 7b: Diffusion large language models. *arXiv preprint arXiv:2508.15487*, 2025.
- You, Z., Nie, S., Zhang, X., Hu, J., Zhou, J., Lu, Z., Wen, J.-R., and Li, C. Llada-v: Large language diffusion models with visual instruction tuning. *arXiv preprint arXiv:2505.16933*, 2025.
- Yue, X., Ni, Y., Zhang, K., Zheng, T., Liu, R., Zhang, G., Stevens, S., Jiang, D., Ren, W., Sun, Y., et al. MMMU: A massive multi-discipline multimodal understanding and

reasoning benchmark for expert AGI. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

Zhang, J., Huang, J., Jin, S., and Lu, S. Vision-language models for vision tasks: A survey. *IEEE transactions on pattern analysis and machine intelligence*, 46(8):5625–5644, 2024a.

Zhang, X., Quan, Y., Gu, C., Shen, C., Yuan, X., Yan, S., Cheng, H., Wu, K., and Ye, J. Seeing clearly by layer two: Enhancing attention heads to alleviate hallucination in LVLMs. *arXiv preprint arXiv:2411.09968*, 2024b.

Zhang, Z., Zhang, A., Li, M., and Smola, A. Automatic chain of thought prompting in large language models. *arXiv preprint arXiv:2210.03493*, 2022.

Zhang, Z., Zhang, A., Li, M., Zhao, H., Karypis, G., and Smola, A. Multimodal chain-of-thought reasoning in language models. *arXiv preprint arXiv:2302.00923*, 2023.

Zheng, G., Yang, B., Tang, J., Zhou, H.-Y., and Yang, S. Dd-cot: Duty-distinct chain-of-thought prompting for multimodal reasoning in language models. *Advances in Neural Information Processing Systems*, 36:5168–5191, 2023.

Table 8. Instance-level flip analysis for LaViDa at $L/T = 256/128$. $W \rightarrow R$ and $R \rightarrow W$ denote examples that change from wrong to right and from right to wrong, respectively.

Benchmark	Base	ST-Veto	Δ	$W \rightarrow R$	$R \rightarrow W$	Flip ratio / Z
MMBench	31.55	38.25	+6.70	518	228	2.27 / 10.62
SQA-IMG	40.06	48.53	+8.47	410	239	1.72 / 6.71

Table 9. Stagewise distribution of Veto-and-Swap events on LaViDa. Values are percentages over all swap events.

Normalized timestep range	64 steps	128 steps
0–17%	18.3	19.6
17–33%	38.3	37.4
33–50%	33.7	33.4
50–67%	8.6	6.9
67–83%	5.4	3.5
83–100%	1.8	1.0

A. Additional Experiments and Analyses

This appendix provides additional evidence discussed during the review process. We focus on the robustness of ST-Veto to generation length, the role of early denoising steps, comparisons with additional benchmarks, and diagnostic analyses showing that the gains are not artifacts of output format or random decoding perturbations.

A.1. Prompt Configurations and Experimental Settings

For all benchmarks, we follow the official question format and answer extraction protocol whenever available. The base dMLLMs are evaluated in their reasoning mode, where the model is allowed to generate an intermediate rationale before producing the final answer. For CoT-style baselines, we use the original prompt templates proposed by each method and keep the same diffusion remasking policy as the base decoder unless the method explicitly requires an additional visual prompting component. This keeps the comparison focused on the reasoning or decoding strategy rather than on prompt-format differences.

All main experiments use deterministic decoding with argmax token selection at each denoising step, without temperature sampling. Unless otherwise specified, Low-confidence remasking is used as the default remasking policy. We evaluate the two generation settings used throughout the paper, $L/T = 128/64$ and $L/T = 256/128$, where L is the maximum generation length and T is the number of denoising steps. Additional long-generation results with $L/T = 512/256$ are reported in Appendix A.4.

A.2. Evaluation Protocol and Output Validity

All reported benchmarks use rule-based or official answer extraction rather than LLM/VLM-as-judge evaluation. Thus, improvements are not caused by stylistic differences in generated rationales. We further verified that both the base decoder and ST-Veto produce zero invalid or unparseable outputs on MMBench and ScienceQA in the settings analyzed below. The gains therefore come from changes in the selected answer option, not from cleaner formatting.

Table 8 reports a paired instance-level flip analysis. If ST-Veto merely introduced random perturbations, the numbers of $W \rightarrow R$ and $R \rightarrow W$ flips would be approximately balanced. Instead, the flip ratios are strongly asymmetric, with large positive Z -scores. We also observe broad category-level consistency: on MMBench, 17 of 20 categories improve; on ScienceQA, 8 of 9 grade levels and all 3 subjects improve. This pattern supports the interpretation that ST-Veto improves token-level selection rather than relying on stochastic answer changes.

A.3. Stagewise Veto-and-Swap Dynamics

Table 9 reports where swaps occur during denoising. The swap ratio is larger in early and middle stages, where token commitments can act as anchors for later refinement. ST-Veto is still useful in late stages, but its largest effect comes from avoiding premature commitment to unstable or weakly grounded tokens before they influence subsequent steps.

Table 10. Longer generation results on LaViDa with $L/T = 512/256$.

Dataset	Base	ST-Veto
MMBench	32.54	38.12
SQA-IMG	41.22	48.72
M ³ CoT	30.58	31.76
V* Bench	36.65	42.20

Table 11. Additional benchmark results on LaViDa with $L/T = 128/64$. MME subtasks report the official score; POPE and MMMU report accuracy.

Dataset	Base	CCoT	DDCoT	ICoT	ST-Veto
MME Exist.	183.33	185.00	176.67	181.67	187.00
MME Count	133.33	126.67	137.22	136.33	143.33
MME Pos.	86.67	90.55	81.67	89.33	91.67
MME Color	141.67	148.33	149.17	147.67	150.00
MME Total	545.00	550.55	544.73	545.00	570.00
POPE	84.45	84.95	85.10	84.95	87.65
MMMU	30.44	31.13	30.54	31.22	33.41

A.4. Generation Length

We additionally test a longer generation setting with $L = 512$ and $T = 256$. ST-Veto maintains its advantage, indicating that the policy does not require a larger rejection ratio as output length grows.

A.5. Additional Benchmarks

We also evaluate LaViDa at $L/T = 128/64$ on MME (Fu et al., 2023), POPE (Li et al., 2023c), and MMMU (Yue et al., 2024). These results use the same reasoning mode as the main experiments; therefore, they are not directly comparable to reports that restrict generation to only a few final-answer tokens.

A.6. Training-Free Decoding Baselines

Most existing hallucination-mitigation decoders are designed for autoregressive MLLMs, so they are not protocol-aligned with diffusion decoding. We nevertheless adapt representative training-free baselines, including EAH (Zhang et al., 2024b), OPERA (Huang et al., 2024), and FarSight (Tang et al., 2025). EAH provides the strongest non-ST-Veto gain, suggesting that encouraging image attention is also helpful for dMLLMs. OPERA and FarSight are less compatible because they rely on autoregressive beam search behavior or causal-mask modification.

A.7. What Tokens Are Vetoed and Swapped?

To better understand the swap behavior, we analyze the part-of-speech distribution of vetoed top- k tokens and accepted swap tokens. Vetoed tokens are dominated by function words, while swapped tokens are more often content words. This is consistent with the early-anchor interpretation: ST-Veto tends to delay unstable high-probability fillers and instead commits to semantically informative tokens when they are temporally stable and visually grounded.

A.8. Fine-Grained Category Analysis

Table 14 reports representative fine-grained MMBench categories. ST-Veto is especially helpful for structured image-text understanding, physical relations, attribute comparison, and recognition tasks. The small number of negative categories indicates that the policy has limited side effects because it does not directly rewrite logits or attention maps.

Table 12. Comparison with additional training-free decoding baselines on LaViDa.

Dataset	L/T	Base	EAH	OPERA	FarSight	ST-Veto
MMBench	128/64	33.96	34.96	34.03	34.23	38.05
	256/128	31.55	33.89	31.92	31.67	38.25
SQA-IMG	128/64	45.81	45.52	45.75	45.22	47.80
	256/128	40.06	43.44	41.12	40.85	48.54
M ³ CoT	128/64	27.96	27.87	28.11	27.88	30.21
	256/128	29.55	29.45	29.56	29.65	31.87
V* Bench	128/64	35.60	37.17	36.65	35.60	42.94
	256/128	37.17	39.27	35.60	36.65	41.36

Table 13. Part-of-speech summary for vetoed and swapped tokens. Percentages report the most frequent groups; remaining categories are grouped as Other.

Vetoed tokens			Swapped tokens		
Part of speech	Ratio	Examples	Part of speech	Ratio	Examples
Determiner	28.4	the, a, an	Noun	29.7	dog, table, color
Preposition	19.1	of, in, at, to	Verb	18.4	holding, placed, sitting
Conjunction	14.6	and, or, but	Preposition	14.2	in, on, between
Other	37.9	–	Other	37.7	–

A.9. Limitations and Future Work

ST-Veto relies on internal confidence and attention signals, which are useful but imperfect proxies. Non-smooth confidence trajectories can make Taylor-based prediction less reliable, and image attention may be diffuse or semantically misaligned with the visual evidence needed for an answer. Our conservative Veto-and-Swap policy mitigates these issues by requiring a safe near-boundary candidate before replacing a token, but it cannot guarantee correction in every case.

The present study focuses on single-image multimodal reasoning with current dMLLMs. Extending the method to multi-image inputs, video understanding, and ultra-long generations will require additional design, particularly for normalizing visual grounding signals across many visual tokens and for handling attention dilution as the generated context grows. We view these settings as important directions for diffusion-native multimodal reasoning.

Table 14. Representative fine-grained MMBench category results.

Category	N	Base	ST-Veto	Diff.
Structuralized image-text understanding	282	36.88	57.09	+20.21
Physical relation	94	34.04	48.94	+14.89
Image topic	140	31.43	44.29	+12.86
Action recognition	215	28.37	40.47	+12.09
Attribute comparison	141	32.62	44.68	+12.06
Celebrity recognition	396	25.51	27.02	+1.52
Spatial relationship	177	28.25	28.81	+0.56
Attribute recognition	264	35.61	35.23	-0.38
Image emotion	200	40.50	40.00	-0.50
Identity reasoning	176	26.14	25.00	-1.14